Ultimate Deoxidation Method of Titanium Utilizing Y/YOCl/YCl_3 Equilibrium

AKIHIRO IIZUKA, TAKANARI OUCHI, and TORU H. OKABE

Deoxidation methods of titanium (Ti) scrap and Ti powder have become increasingly important in recent years. Some rare earth (RE) metals with strong deoxidizing capabilities, including Y, La, Ce, and Ho, are candidate agents for the development of a new deoxidation technology. In this study, a new method was developed to directly remove oxygen (O) from Ti using the Y/YOCl/YCl_3 equilibrium. According to the calculation based on available thermodynamic data in the literature, the O concentration in β-Ti can be reduced to less than 10 ppm O at 1300 K (1027 °C) in the Y/YOCl/YCl_3 equilibrium. To demonstrate the effectiveness of this method using the Y/YOCl/YCl_3 equilibrium, the deoxidation limits of Ti samples using Y metal in YCl_3 (*l*) or in YCl_3-NaCl-KCl (*l*) at 1300 K (1027 °C) were experimentally investigated in this study. As a result, the O concentrations in the Ti samples were from 30 to 60 ppm O in YCl_3 (*l*). This result revealed that Ti with extremely low O concentration can be reliably obtained using the RE/REOCl/$RECl_3$ equilibrium for the first time. The establishment of this process will realize efficient recycling of Ti scrap and production of low O concentration Ti powder, which contribute to the large-scale use of Ti products.

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I. INTRODUCTION

TITANIUM (Ti) has a strong affinity for oxygen (O), and the solubility of O in Ti is high.[1–3] Because O concentration in Ti drastically changes the properties of Ti, various methods have been used industrially to suppress O contamination in smelting, machining, and working processes. As Ti products have become widespread particularly in the aerospace industry, deoxidation methods of Ti scrap and Ti powder have become increasingly important in recent years.[4]

There was no effective method for directly removing O in Ti until the 1990s because of the strong reactivity of Ti. Since then, through the use of molten salt, the method using metal/metal oxide equilibrium (Figures 1(a) and (d)), the calcium-halide flux method (Figure 1(b)), and the electrochemical deoxidation method (Figure 1(c)) have been actively developed. However, large-scale industrialized direct removal of O from Ti remains challenging. Therefore, we focus on the development of new methods that can reduce the O concentration in Ti more simply and reliably, rather than just the modification of existing methods.

In general, calcium (Ca) and rare earth (RE) metals, which have a strong affinity for O, are used in thermochemical deoxidation methods for Ti in metal/metal oxide equilibrium. In particular, methods to deoxidize Ti in Ca/CaO equilibrium have been widely studied (Figure 1(a)).[5–22] For example, when the activity of each chemical species is unity ($a_{Ca} = 1, a_{CaO} = 1$) at 1300 K (1027 °C), O in Ti can be removed to approximately 500 ppm O in the Ca/CaO equilibrium. Additionally, the RMI Titanium company applied this deoxidation method using Ca vapor, referred to as the "deoxidation in solid state" (DOSS) method, on an industrial scale.[18–22] Until recently, many researchers have actively developed deoxidation techniques for Ti bulk or Ti powder using Ca metal.[8–17]

In 1992, Okabe *et al.* demonstrated that O concentration in Ti can be dramatically reduced to 20 ppm O by decreasing the activity of CaO, a_{CaO}, in $CaCl_2$ flux, which has a large solubility of CaO (calcium-halide flux method, Figure 1(b)).[23–26] Recently, the calcium-halide flux method has been studied by Han *et al.* and Xia *et al.*[27,28] However, there is no cost-effective process for regenerating Ca metal from CaO produced in the calcium-halide flux method.

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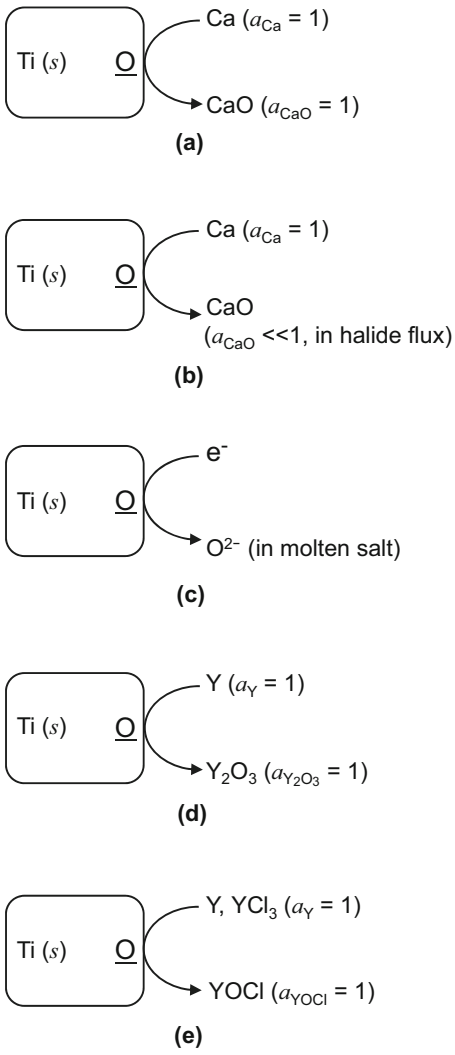


Fig. 1—Deoxidation techniques of Ti in molten salt. (*a*) Deoxidation of Ti in the Ca/CaO equilibrium. (*b*) Deoxidation of Ti in calcium-halide flux. (*c*) Electrochemical deoxidation of Ti. (*d*) Deoxidation of Ti in the Y/Y_2O_3 equilibrium. (*e*) Deoxidation of Ti by oxyhalide formation.

In 1993, Okabe *et al.* proposed an electrochemical method to directly remove O from Ti by applying voltage between the Ti cathode and the carbon anode in $CaCl_2$ flux.[29] They experimentally demonstrated that the O concentration in Ti can be reduced to lower than 10 ppm O using this method. Using a similar approach, Nakamura *et al.* and Taninouchi *et al.* demonstrated that O concentration in Ti can be decreased to 100 ppm O or less in $MgCl_2$ or $BaCl_2$ flux.[30–34] In 2000, Chen *et al.* applied the electrochemical deoxidation method to direct reduction of TiO_2. They reduced TiO_2 powder to pure Ti with a level of 200 ppm O or less.[35] This method is known as the Fray–Farthing–Chen Cambridge (FFC) method, and has been extensively researched as a reduction method for many reactive metals.[36–41]

Furthermore, Okabe *et al.* also reported the deoxidizing capability of the rare earth metal/rare earth metal oxide (RE/RE_2O_3) equilibrium to Ti.[42–45] In particular, it was reported that Y has the strongest deoxidizing capability, and that O concentration in Ti can be decreased to a level of 80 to 220 ppm O in the Y/Y_2O_3 equilibrium when $a_Y = 1$ and $a_{Y_2O_3} = 1$ at 1213 K (940 °C) (Figure 1(d)).

In addition to these Ti deoxidation studies, our group has recently shown the possibility of direct removal of O from Ti with the formation of oxychloride or oxyfluoride of some RE metals (Y, La, Ce, Pr, Nd, and Ho) through thermodynamic considerations.[46–51] For example, in $Y/YOCl/YCl_3$ equilibrium at 1300 K (1027 °C), the O level in Ti can be theoretically reduced to 10 ppm or less (Figure 1(e)). However, to the best of our knowledge, experimental data regarding the deoxidation method of Ti using this equilibrium have not been reported.

During the last 30 years, the production of RE metals has increased to meet the demand for RE alloy magnets which use Nd and Dy as key elements.[52,53] In the future, an oversupply of RE metals such as Y, La, Ce, and Ho is expected because they are by-products of the production of Nd and Dy. Therefore, the development of new industrial applications for Y, La, Ce and Ho metals is highly desirable.

In this study, we have experimentally verified the new deoxidation method using the $Y/YOCl/YCl_3$ equilibrium, which can be expected to have the highest deoxidizing capability among the predictions in the thermodynamic consideration (Figure 1(e)). The thermodynamic data used for the calculation may contain large errors, and the effectiveness of the deoxidation method has not yet been experimentally demonstrated. To realize a new deoxidation technique under equilibrium conditions, an experimental investigation was conducted.

II. DETERMINATION OF DEOXIDATION REACTION AND MOLTEN SALT BASED ON THERMODYNAMIC CONSIDERATIONS

In the Y-O-Cl system at 1300 K (1027 °C), Y_2O_3 (*s*), YOCl (*s*), and YCl_3 (*l*) have been reported as stable compounds.[54,55] The relationship between the standard Gibbs energy of formation of each compound, $\Delta G^\circ_{f,\,i}$, and the activity, a_i, is expressed as follows:

$$2\ \mathrm{Y}\ (s) + 3/2\ \mathrm{O_2}\ (g) = \mathrm{Y_2O_3}\ (s) \quad [1]$$

$$\Delta G^{\circ}_{\mathrm{f,\ Y_2O_3}} = -\ RT\ \ln(a_{\mathrm{Y_2O_3}}/(a_{\mathrm{Y}}{}^2 \cdot p_{\mathrm{O_2}}{}^{3/2})) \quad [2]$$

$$\mathrm{Y}\ (s) + 1/2\ \mathrm{O_2}\ (g) + 1/2\ \mathrm{Cl_2}\ (g) = \mathrm{YOCl}\ (s) \quad [3]$$

$$\Delta G^{\circ}_{\mathrm{f,\ YOCl}} = -\ RT \ln\left(a_{\mathrm{YOCl}}/\left(a_{\mathrm{Y}} \cdot p_{\mathrm{O_2}}{}^{1/2} \cdot p_{\mathrm{Cl_2}}{}^{1/2}\right)\right) \quad [4]$$

$$\mathrm{Y}\ (s) + 3/2\ \mathrm{Cl_2}\ (g) = \mathrm{YCl_3}\ (l) \quad [5]$$

$$\Delta G^{\circ}_{\mathrm{f,\ YCl_3}} = -\ RT \ln\left(a_{\mathrm{YCl_3}}/\left(a_{\mathrm{Y}} \cdot p_{\mathrm{Cl_2}}{}^{3/2}\right)\right), \quad [6]$$

where R is the gas constant and T is absolute temperature.

In Y/Y_2O_3 equilibrium or $Y/YOCl/YCl_3$ equilibrium, assuming that the activities of the stable chemical species of the Y-O-Cl system at 1300 K (1027 °C) are unity, the partial pressure of O_2 (O_2 potential), p_{O_2}, is determined based on equilibrium theory. The relationship between the concentration of O in β-Ti ($[O]_{Ti}$ (mass pct)) and p_{O_2} at 1173 to 1373 K (900 to 1100 °C) can be expressed as follows[7]:

$$1/2\ \mathrm{O_2}\ (g) = \mathrm{O}\ (1\ \text{mass pct},\ \text{in}\ \beta\text{-Ti}) \quad [7]$$

$$\Delta G^{\circ}_{1,\ \mathrm{Ti}} = -2.303\ RT \log(f_{\mathrm{O}} \cdot [\mathrm{O}]_{\mathrm{Ti}}/p_{\mathrm{O_2}}{}^{1/2}) \quad [8]$$

$$\Delta G^{\circ}_{1,\ \mathrm{Ti}} = -583000 + 88.5T\ (\mathrm{J})\ [1173{-}1373\ \mathrm{K}], \quad [9]$$

where O dissolved in Ti is expressed relative to the 1 mass pct standard state, $\Delta G^{\circ}_{1,\ Ti}$ is the standard Gibbs energy of O dissolution in Ti, $[O]_{Ti}$ is the concentration of O in β-Ti (mass pct O), and f_O is the Henrian activity coefficient for O dissolved in Ti. Given that the O dissolved in Ti obeys Henry's law, the value of f_O is considered to be unity.

Because YCl_3 is highly hydroscopic, high-purity anhydrous YCl_3 is expensive as a flux. When using YCl_3 in the experiments, it is desirable to use YCl_3 in combination with other inexpensive salts. Therefore, chemical equilibria between molten salts (LiCl, NaCl, KCl, $CaCl_2$, and $MgCl_2$) and chemical species consisting of Y (Y (*s*), Y_2O_3 (*s*), YOCl (*s*), and YCl_3 (*l*)) were considered.

The thermodynamic relationship between these chemical species was analyzed by superimposing the p_{Cl_2}-p_{O_2} diagram of the M-O-Cl system (M = Li, Na, K, Ca, and Mg) on the p_{Cl_2}-p_{O_2} diagram of the Y-O-Cl system (Figure 2). Table I shows the thermodynamic data used to draw Figure 2. As can be seen from Figure 2, LiCl, NaCl, KCl, and $CaCl_2$ have regions coexisting with the

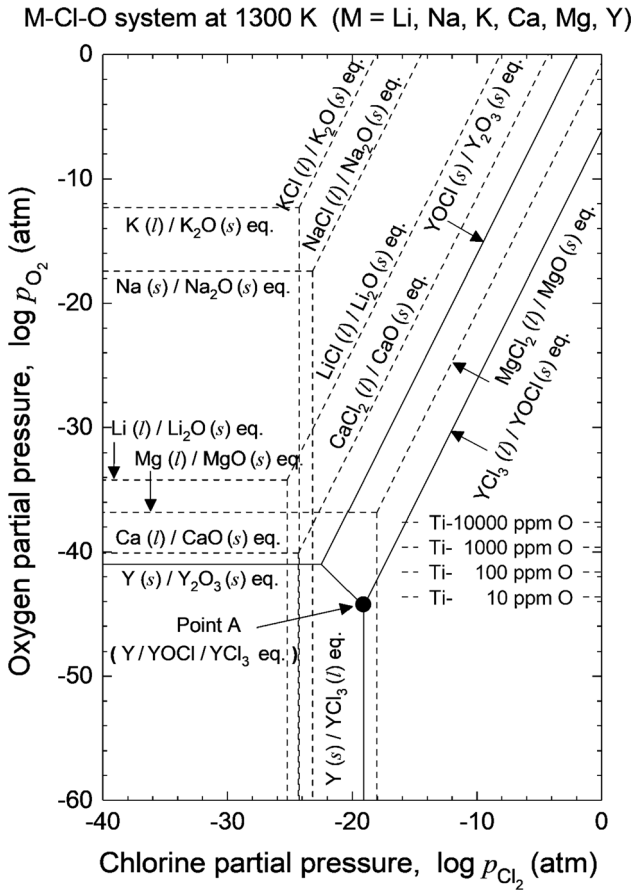


Fig. 2—Phase stability diagram of the M-Cl-O systems (M = Li, Na, K, Mg, Ca, and Y) at 1300 K (1027 °C)[7,54,55]

Table I. Standard Gibbs Energy of Formation at 1300 K Used to Draw Figs. 2 and 3[7,54,55]

Species, *i*	Standard Gibbs energy of formation, $\Delta G^{\circ}_{f,i}$/kJ · mol^{-1}	References
O (1 mass pct, in β-Ti)	− 468	7
LiCl (*l*)	− 314	54
Li_2O (*s*)	− 426	54
NaCl (*l*)	− 288	54
Na_2O (*s*)	− 216	54
KCl (*l*)	− 302	54
K_2O (*l*)	− 153	54
$MgCl_2$ (*l*)	− 449	54
MgO (*s*)	− 458	54
$CaCl_2$ (*l*)	− 605	54
CaO (*s*)	− 499	54
Y_2O_3 (*s*)	− 1530	54
YCl_3 (*l*)	− 713	54
YOCl (*s*)	− 790	55

Y/Y_2O_3 equilibrium and the $Y/YOCl/YCl_3$ equilibrium (Point A). That is, in LiCl, NaCl, KCl, and $CaCl_2$, Y metal is thermodynamically stable, and the following deoxidation reactions proceed at 1300 K (1027 °C):

$$O\ (\text{in Ti}) + 2/3\ Y\ (s) = 1/3\ Y_2O_3\ (s) \quad [10]$$

$$\Delta G^{\circ}_{r,\ deox[10]} = 1/3\ \Delta G^{\circ}_{f,Y_2O_3} - \Delta G^{\circ}_{1,Ti} \quad [11]$$

$$[O]_{Ti[10]} = \left(a_{Y_2O_3}{}^{1/3}/a_Y{}^{2/3}\right)\exp\left(\Delta G^{\circ}_{r,\ deox}/\ RT\right) \quad [12]$$

$$C_{O,\ theo[10]} = 200\left(a_{Y_2O_3}{}^{1/3}/a_Y{}^{2/3}\right)\ [\text{mass ppm O}] \quad [13]$$

$$O\ (\text{in Ti}) + 2/3\ Y\ (s) + 1/3\ YCl_3\ (l) = \ YOCl\ (s) \quad [14]$$

$$\Delta G^{\circ}_{r,\ deox[14]} = \Delta G^{\circ}_{f,YOCl} - 1/3\ \Delta G^{\circ}_{f,YCl_3} - \Delta G^{\circ}_{1,Ti} \quad [15]$$

$$[O]_{Ti[14]} = \left(a_{YOCl}/\left(a_Y{}^{2/3}\cdot a_{YCl_3}{}^{1/3}\right)\right)\exp\left(\Delta G^{\circ}_{r,deox}/\ RT\right) \quad [16]$$

$$C_{O,\ theo[14]} = 3.4\left(a_{YOCl}/\left(a_Y{}^{2/3}\cdot a_{YCl_3}{}^{1/3}\right)\right)\ [\text{mass ppm O}], \quad [17]$$

where $C_{O,\ theo}$ is the concentration of O in β-Ti (mass ppm O).

Thermodynamic data, such as the standard Gibbs energy of formation of YOCl, may contain large errors. The errors in thermodynamic data used in the calculation drastically affect the calculated deoxidation limit. For example, if the standard Gibbs energy of the deoxidation reaction has $\pm$ 10 kJ·mol^{-1} error, log $[O]_{Ti}$ changes by approximately $\pm$ 0.4.

Conversely, the following displacement reaction occurs when Y metal is placed in $MgCl_2$.

$$3\ MgCl_2\ (l) + 2\ Y\ (s) \rightarrow 2\ YCl_3\ (l) + 3\ Mg\ (l) \quad [18]$$

As long as $MgCl_2$ exists in the system, Y metal induces the displacement reaction (Eq. [18]) with $MgCl_2$ and disappears in the system. Then, the deoxidation limit of Ti is determined by the $Mg/MgCl_2/YOCl/YCl_3$ equilibrium. Our group recently reported the deoxidation methods in such equilibria (*e.g.*, $Mg/MgCl_2/YOCl/YCl_3$ equilibrium,[46,47] $Mg/MgCl_2/HoOCl/HoCl_3$ equilibrium.[48,49]) Therefore, $MgCl_2$ was excluded from the candidates of the solvent in this study because the $Y/YOCl/YCl_3$ equilibrium cannot be achieved in the system containing $MgCl_2$, and the deoxidizing capability under the equilibrium cannot be verified directly in $MgCl_2$.

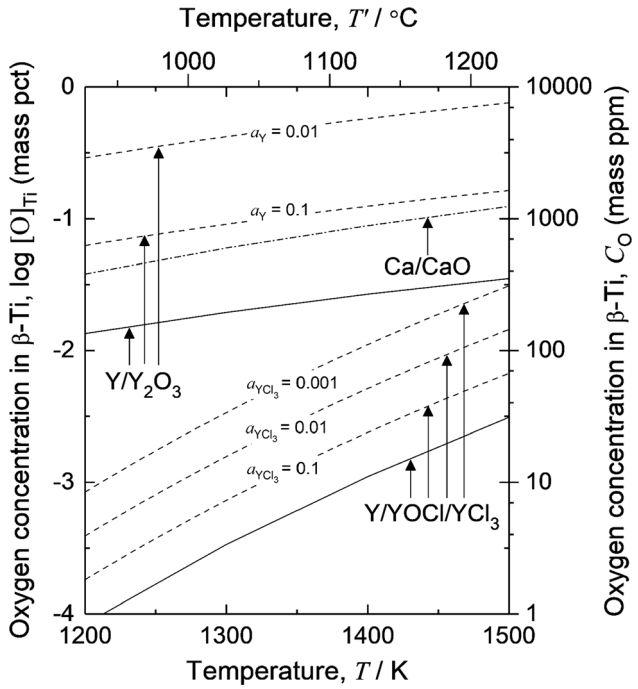


Fig. 3—Temperature dependence of oxygen concentration in β-Ti determined by the M/M_2O_3 and $M/MOCl/MCl_3$ (M = Y, Ca) equilibria, with varying a_Y and a_{YCl_3}[7,54,55]

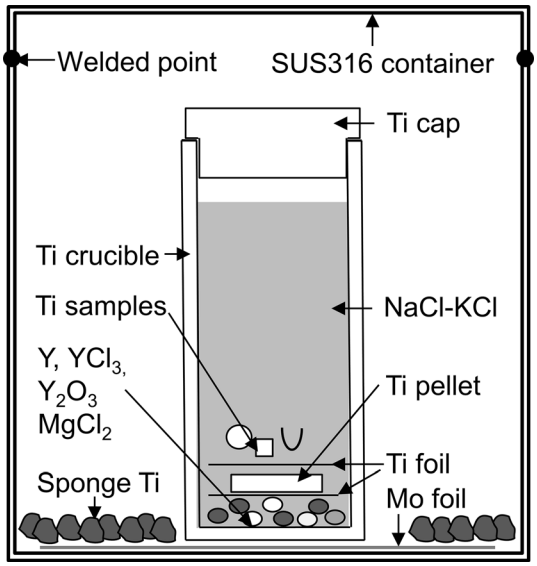


Fig. 4—Schematic illustration of the experiment.

Based on the considerations above, in this study, we used YCl_3 as a molten salt mixed with and without the eutectic NaCl-KCl (50 mol pct NaCl-KCl). The calculated deoxidation limit of Ti using available thermodynamic data in the literature at 1300 K (1027 °C) is 200 ppm O in the Y/Y_2O_3 equilibrium and 3.4 ppm O in the $Y/YOCl/YCl_3$ equilibrium (Point A, Figure 2), when

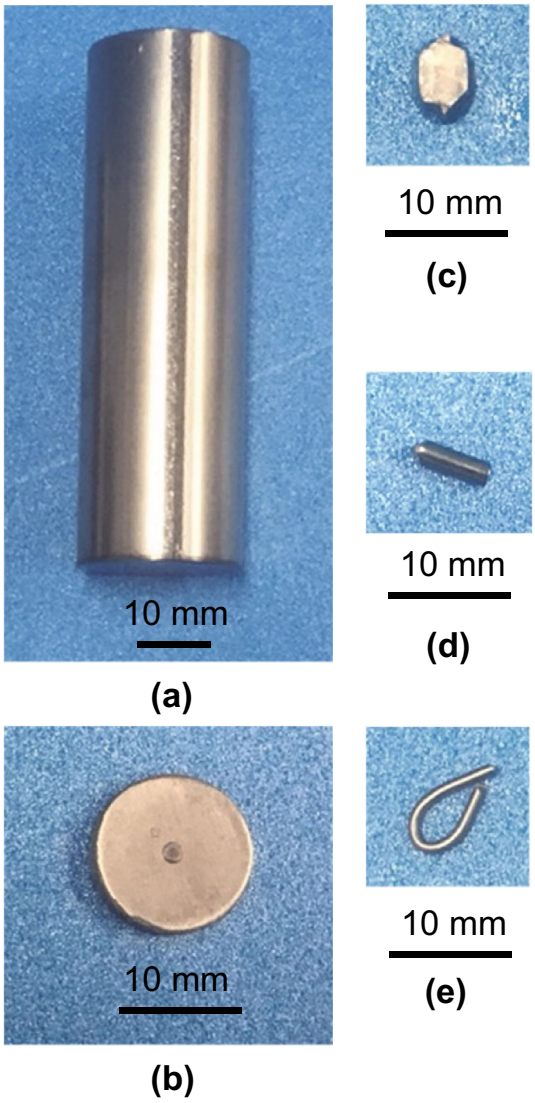


Fig. 5—Photographs of experimental apparatuses: (*a*) Ti crucible (25.4 mm in diameter), (*b*) Ti pellet, (*c*) Ti sample (2B, 2 mm Ti squared piece), (*d*) Ti sample (2L, Ti wire, diameter: 2 mm), and (*e*) Ti sample (1.2U, Ti wire, diameter: 1.2 mm).

the activities of the stable chemical species are unity. Furthermore, Figure 3 shows the deoxidation limit in Ti when the activities of Y and YCl_3 change. As shown in this figure, when the activity of YCl_3 is 0.1, the deoxidation limit in Ti in the Y/YOCl/YCl_3 equilibrium is 7.3 ppm O. Based on the thermodynamic calculation, the O concentration in Ti can be decreased to an extremely low level in the molten salt, even in YCl_3 mixed into NaCl-KCl, when using the deoxidizing capability of Y metal.

III. EXPERIMENTAL PROCEDURE

Figure 4 shows a schematic of the experimental apparatus to achieve the Y/YOCl/YCl_3 equilibrium. The Ti crucible was placed with sponge Ti (getter) in a stainless steel container which is completely sealed by tungsten inert gas welding. Y metal was placed on the bottom of the Ti crucible. Approximately 0.1-g Ti samples used to measure O concentration were set in the crucible as shown in Figure 4. The Ti samples used in this study were cubes with 2 to 3 mm sides (Ti-2B, *ca.* 230 ppm O), a wire with 1.2 mm diameter (Ti-1.2U, *ca.* 1200 ppm O), and a wire with 2 mm diameter (Ti-2L, *ca.* 1200 ppm O). Photographs of the experimental apparatus are shown in Figure 5, and detailed information of samples and reagents used in the experiments is shown in Table II.

The initial amounts of samples and reagents in each lot are shown in Table III. In Exps. #181001-lot.3-6, #181001-lot.3-7, #181212-lot.4-6, and #181212-lot.4-7, YCl_3 is produced by the displacement reaction between $MgCl_2$ and Y metal (Eq. (18)). In order to consume all $MgCl_2$ in the system, an excess amount of Y metal was placed in the Ti crucible. In Exp. #190315-lot.6-1, anhydrous YCl_3 was initially placed in the Ti crucible.

Moreover, the experiment under the Y/Y_2O_3 equilibrium was also conducted as a comparative experiment. Detailed information of samples used in the experiments is shown in Table II, and the initial amounts of samples and reagents in each lot are shown in Table IV. The procedure was almost the same as in the case of the experiment with the Y/YOCl/YCl_3 equilibrium. Considering the solubility of Y_2O_3 in the molten salt, approximately 1 g of Y_2O_3 was added from before the experiment to perform the reaction to maintain $a_{Y_2O_3}$, at unity.

The stainless steel container was placed in a box-type electric furnace and heated to 1300 K (1027 °C). The holding time at 1300 K (1027 °C) was determined based on the diffusion coefficient of O in Ti at 1300 K (1027 °C),[56] and the shapes of Ti samples. After holding the stainless steel container at 1300 K (1027 °C) for 173 ks, the container was quenched in water. The Ti samples were chemically polished with a 1:4:10 mixture of HF-HNO_3-H_2O, and then rinsed with distilled water and alcohol before drying. The concentration of O and nitrogen (N) in the Ti sample was analyzed by the inert gas fusion technique (TC-600, LECO Corporation). Analysis conditions are shown in Tables V and VI in detail.

Phase identification of salts in the Ti crucible after the experiments was conducted using an X-ray diffraction (XRD) technique (Bruker, D2 Phaser, Cu-Kα radiation).

IV. RESULTS AND DISCUSSION

Tables V and VI show the O and N concentrations in Ti samples after the experiment. As shown in Table V, when pure YCl_3 was employed as the molten salt (Exps. #181001-lot.3-6, #181212-lot.4-6, #190315-lot.6-1), the O concentrations of Ti samples after the experiment were approximately 30 to 60 ppm O. When 10 mol pct YCl_3-NaCl-KCl was employed as the molten salt (Exps. #181001-lot.3-7, #181212-lot.4-7), the O concentrations of Ti samples were approximately 80 to 100 ppm O. In the comparative experiments to achieve the Y/Y_2O_3 equilibrium, the O concentrations in Ti (Exps.

Table II. Materials and Regents Used in the Experiments

Materials	Form	Purity/Grade	Supplier/Note
Ti crucible	25.4 mm in diameter 1 mm in thickness 80 mm in height	CP-Ti[a]	Shinkinzoku Industry Co., Ltd.
Ti cap		CP-Ti[a]	Shinkinzoku Industry Co., Ltd.
Ti foil	0.1 mm in thickness	CP-Ti[a]	Shinkinzoku Industry Co., Ltd.
Ti-2B	Cube with 2 mm in length of side	~ 230 mass ppm O	High-purity Ti sponge, electron-beam melted
Ti-1.2U	Wire with 1.2 mm in diameter	~ 1200 mass ppm O	Shinkinzoku Industry Co., Ltd.
Ti-2L	Wire with 2 mm in diameter	~ 1200 mass ppm O	Shinkinzoku Industry Co., Ltd.
Ti	Sponge	≥ 97.0 pct	Toho Titanium Co., Ltd.
Ti	Pellet made of HDH powder, (particle size < 45 μm)	≥ 99.4 pct < 3500 ppm O < 300 ppm N	Osaka Titanium Technologies Co., Ltd.
Y	Shot	≥ 99 pct	Santoku Co., Ltd.
$MgCl_2$	Powder	≥ 97 pct, anhydrous	Wako Industry Co., Ltd.
YCl_3	Powder	≥ 99.9 pct, anhydrous	Alfa Aesar Co., Ltd.
Y_2O_3	Powder	≥ 99.9 pct	Shin-Etsu Chemical Co., Ltd.
NaCl	Powder	≥ 99 pct	Wako Industry Co., Ltd.
KCl	Powder	≥ 99 pct	Wako Industry Co., Ltd.
Stainless steel crucible (tube and plate)		SUS 316, Ni: 10–14 pct Cr: 16–18 pct Mo: 2–3 pct Fe: balanced	Azabu Industry Co., Ltd.

[a]Commercially pure titanium (~ 99.5 pct).

Table III. Initial Amounts of Samples and Reagents in the Ti Crucible for the Experiment Utilizing the Y/YOCl/YCl_3 Equilibrium

Exp. No.,	Activity of Y, a_Y	Mole fraction of YCl_3, X_{YCl_3}[a]	Initial amount of samples in the titanium crucible, m/g (n/mol)					Theoretical deoxidation limit of Ti, $C_{O,\,theo}$ (mass ppm)[c]
			NaCl	KCl	Y	$MgCl_2$	YCl_3	
181001-lot.3-6[b]	1	1			13.33 (0.150)	11.43 (0.120)		3.4
181212-lot.4-6[b]	1	1			13.34 (0.150)	11.43 (0.120)		3.4
190315-lot.6-1	1	1			2.83 (0.0318)		21.69 (0.111)	3.4
181001-lot.3-7[b]	1	0.1	5.26 (0.0900)	6.71 (0.0900)	8.89 (0.100)	2.86 (0.0300)		7.3
181212-lot.4-7[b]	1	0.1	5.26 (0.0900)	6.71 (0.0900)	8.89 (0.100)	2.86 (0.0300)		7.3

[a]Calculated from the initial amount of the samples (coefficient of YCl_3 is unity).
[b]$Y + 3/2\ MgCl_2 \rightarrow YCl_3 + 3/2\ Mg$ was occurred.
[c]$a_Y = 1$, $a_{YOCl} = 1$, $a_{YCl_3} = n_{YCl_3}/(n_{YCl_3} + n_{NaCl} + n_{KCl})$.

Table IV. Initial Amounts of Samples and Reagents in the Ti Crucible for the Experiment Utilizing the Y/Y_2O_3 Equilibrium

Exp. No.,	Activity of Y, a_Y	Initial amount of samples in the titanium crucible, m/g (n/mol)				Theoretical deoxidation limit of Ti, $C_{O,\,theo}$ (mass ppm)
		NaCl	KCl	Y	Y_2O_3	
181001-lot.3-1	1	16.22 (0.278)	20.68 (0.278)	4.42 (0.0497)	1.13 (0.00500)	200
181212-lot.4-1	1	14.72 (0.252)	18.78 (0.252)	5.33 (0.0599)	1.16 (0.00514)	200

Table V. Results of Oxygen and Nitrogen Analyses of Ti Samples in the Experiment Utilizing the Y/YOCl/YCl_3 Equilibrium

Exp. No.	Molten salt	Types of Ti samples[a]	Oxygen concentration, C_O (mass ppm)		Analyzed error, ε_O (pct)	Nitrogen concentration, C_N (mass ppm)		Analyzed error, ε_N (pct)	Theoretical deoxidation limit of Ti, $C_{O,\,theo}$ (mass ppm)[f]
			Initial, $C_{O,\,initial}$	After exp., $C_{O,\,after}$		Initial, $C_{N,\,initial}$	After exp., $C_{N,\,after}$		
181001-lot.3-6[b]	YCl_3[e]	2L *2 pieces	1200	50	51	30	2200	10	3.4
		2B	230	140[g]	14	20	2000	10	
		2B	230	40	61	20	2100	10	
181212-lot.4-6[c]	YCl_3[e]	2L	1200	60	36	30	2300	12	3.4
		1.2U	1200	40	55	30	2600	12	
		2B	230	50	26	20	2000	12	
190315-lot.6-1[d]	YCl_3	2L	1200	30	20	30	140	14	3.4
		2L	1200	30	22	30	160	14	
		2L	1200	30	22	30	130	17	
		2L	1200	90[g]	7	30	130	14	
		2L	1200	30	20	30	170	14	
		1.2U	1200	40	19	30	190	14	
		2B	230	60	14	30	140	16	
181001-lot.3-7[b]	10 mol pct YCl_3-NaCl-KCl[e]	2L	1200	80	35	30	500	10	7.3
		2L	1200	100	28	30	330	10	
		2B	230	100	13	20	440	10	
181212-lot.4-7[c]	10 mol pct YCl_3-NaCl-KCl[e]	2L	1200	100	23	30	150	12	7.3
		2L	1200	90	25	30	340	12	
		1.2U	1200	100	24	30	240	12	
		1.2U	1200	90	25	30	210	12	
		2B	230	90	14	20	60	12	

[a]Ti samples: ~ 1.0 g, 2L: wire with 2 mm in diameter, 1.2U: wire with 1.2 mm in diameter, 2B: cube with side length of 2 mm.

[b]Analytical conditions: Standard sample: steel pin ~ 1.0 g, 365±7 ppm O, 22±2 ppm N, part number: 501-646, Graphite crucible: ~ 1.0 g, 0.1±0.1 μg O, 0.0±0.0 μg N, part number: 782-720, Ni flux: ~1.0 g, 4.4±1.9 μg O, 0.0±0.0 μg N, part number: 502-344, Estimated maximum analytical error: $\varepsilon_{O,\,max}$ = 70 pct, $\varepsilon_{N,\,max}$ = 10 pct.

[c]Analytical conditions: Standard sample: steel pin ~ 1.0 g, 365±7 ppm O, 22±2 ppm N, part number: 501-646, Graphite crucible: ~ 1.0 g, 0.0±0.1 μg O, 0.0±0.0 μg N, part number: 782-720, Ni flux: ~1.0 g, 3.4±1.5 μg O, 0.0±0.0 μg N, part number: 502-344, Estimated maximum analytical error: $\varepsilon_{O,\,max}$ = 60 pct, $\varepsilon_{N,\,max}$ = 12 pct.

[d]Analytical conditions: Standard sample: steel pin ~ 1.0 g, 366±6 ppm O, 23±2 ppm N, part number: 502-874, Graphite crucible: ~ 1.0 g, 0.0±0.1 μg O, 0.0±0.0 μg N, part number: 782-720, Ni flux: ~1.0 g, 2.4±0.5 μg O, 0.4±0.8 μg N, part number: 502-344, Estimated maximum analytical error: $\varepsilon_{O,\,max}$ = 30 pct, $\varepsilon_{N,\,max}$ = 20 pct.

[e]$Y + 3/2\ MgCl_2 \rightarrow YCl_3 + 3/2\ Mg$ occurred.

[f]$a_Y = 1$, $a_{YOCl} = 1$, $a_{YCl_3} = n_{YCl_3}/(n_{YCl_3} + n_{NaCl} + n_{KCl})$.

[g]This value deviates from other values and is not reflected in the discussion.

Table VI. Results of Oxygen and Nitrogen Analyses of Ti Samples in the Experiment Utilizing the Y/Y_2O_3 Equilibrium

Exp. No.	Activity of Y, a_Y	Types of Ti samples[a]	Oxygen concentration, C_O (mass ppm)		Analyzed error, ε_O (pct)	Nitrogen concentration, C_N (mass ppm)		Analyzed error, ε_N (pct)	Theoretical deoxidation limit of Ti, $C_{O,\ theo}$ (mass ppm)
			Initial, $C_{O,\ initial}$	After exp., $C_{O,\ after}$		Initial, $C_{N,\ initial}$	After exp., $C_{N,\ after}$		
181001-lot.3-1[b]	1	2L *2 pieces	1200	120	21	30	80	10	200
		2B	230	120	11	20	30	10	
		2B	230	150	11	20	30	10	
181212-lot.4-1[c]	1	2L	1200	210	11	30	80	12	200
		1.2U	1200	80[d]	23	30	40	12	
		2B	230	140	14	20	50	12	

[a]Ti samples: ~ 1.0 g, 2L: wire with 2 mm in diameter, 1.2U: wire with 1.2 mm in diameter, 2B: cube with side length of 2 mm.
[b]Analytical conditions: Standard sample: steel pin ~ 1.0 g, 365±7 ppm O, 22±2 ppm N, part number: 501-646, Graphite crucible: ~ 1.0 g, 0.1±0.1 μg O, 0.0±0.0 μg N, part number: 782-720, Ni flux: ~1.0 g, 4.4±1.9 μg O, 0.0±0.0 μg N, part number: 502-344, Estimated maximum analytical error: $\varepsilon_{O,\ max}$ = 70 pct, $\varepsilon_{N,\ max}$ = 10 pct.
[c]Analytical conditions: Standard sample: steel pin ~ 1.0 g, 365±7 ppm O, 22±2 ppm N, part number: 501-646, Graphite crucible: ~ 1.0 g, 0.0±0.1 μg O, 0.0±0.0 μg N, part number: 782-720, Ni flux: ~1.0 g, 3.4±1.5 μg O, 0.0±0.0 μg N, part number: 502-344, Estimated maximum analytical error: $\varepsilon_{O,\ max}$ = 60 pct, $\varepsilon_{N,\ max}$ = 12 pct.
[d]This value deviates from other values and is not reflected in the discussion.

#181001-lot.3-1, #181212-lot.4-1) were approximately 140 to 210 ppm O (Table VI). The O concentration in Ti in the $Y/YOCl/YCl_3$ equilibrium is apparently lower than that in the Y/Y_2O_3 equilibrium. These results indicate the high effectiveness of the $Y/YOCl/YCl_3$ equilibrium, which can decrease the O concentration in Ti samples to lower than 100 ppm O. The results experimentally demonstrated that this deoxidation method can change impure Ti scrap to pure Ti, the O concentration of which is much lower (30 to 60 ppm O) than that of the virgin material (*e.g.*, Ti sponge: 500 ppm O or less) produced in the Kroll process.

Figure 6 shows the XRD spectrum of salts after the experiments.[57–63] When YCl_3 is produced by the displacement reaction (Eq. [18]) between Y and $MgCl_2$ (Exps. #181001-lot.3-6, #181001-lot. 3-7), no peak attributed to $MgCl_2$ was observed, and the peaks of YCl_3 and $YCl_3{\cdot}6H_2O$ were observed. $YCl_3{\cdot}6H_2O$ was produced when YCl_3 absorbed water in the atmosphere after the experiment. In the experiment when YCl_3 was initially placed (Exp. #190315-lot.6-1), the existence of YCl_3 and $YCl_3{\cdot}6H_2O$ compounds was also confirmed after the experiment. Considering the p_{Cl_2}-p_{O_2} diagram of the Y-O-Cl system shown in Figure 2, when Y and YCl_3 coexist, Y_2O_3 is not stable in the system. These results indicate that the deoxidation reaction (Eq. [14]) occurred, and the system was equilibrated at the $Y/YOCl/YCl_3$ equilibrium in both types of experiments. The peak of YOCl was not clearly observed because the amount of produced YOCl was small for the detection limit in XRD analysis.

The O concentration obtained by the experiment in the $Y/YOCl/YCl_3$ equilibrium (30 to 60 ppm O) is much higher than the calculated value (3.4 ppm O). The difference may come from the following reasons: (1) the difference in the activity of Y, a_Y, which was assumed to be unity in the calculation, and (2) the inaccuracy of the standard Gibbs energy of formation ($\Delta G^\circ_{f,\ YOCl}$ and $\Delta G^\circ_{f,\ YCl_3}$) in the literature.

The Mg metal, which was simultaneously produced when the displacement reaction (Eq. [18]) occurred to produce YCl_3, was likely to reduce a_Y in the system. However, the difference between the results of the experiments using the displacement reaction (Exps. #181001-lot.3-6, #181215-lot.4-6; 40 to 60 ppm O) and those of the experiment directly charged with YCl_3 (Exp. #190315-lot.6-1; 30 to 60 ppm O) was small. This indicates that the effect of decrease in a_Y is small for the deoxidation limit in this study.

In contrast, the error in the thermodynamic data used in the calculation drastically affects the calculated deoxidation limit. When the standard Gibbs energy of the reaction [14], $\Delta G^\circ_{r,\ deox\ [14]}$, is + 25 $kJ{\cdot}mol^{-1}$ higher, the calculated deoxidation limit ($C_{O,\ theo\ [14]}$) changes from 3.4 ppm O to 34 ppm O, which is consistent with the experimental results. Consequently, the inaccuracy of $\Delta G^\circ_{f,\ YOCl}$ and $\Delta G^\circ_{f,\ YCl_3}$ is the main reason for the difference between the calculated and the experimental values. In addition, if YOCl was not saturated in the molten salt, there is a possibility that the error in the thermodynamic data would be larger.

Furthermore, when YCl_3 was produced by the displacement reaction between Y and $MgCl_2$, the N concentration in Ti samples significantly increased after the experiments. This is because the Ti samples were contaminated with N in Y metal during the displacement reaction. In the experiment when YCl_3 was employed initially (Exp. #190315-lot.6-1), the N concentration was low, which also indicates that Y metal is the source of N contamination. However, N

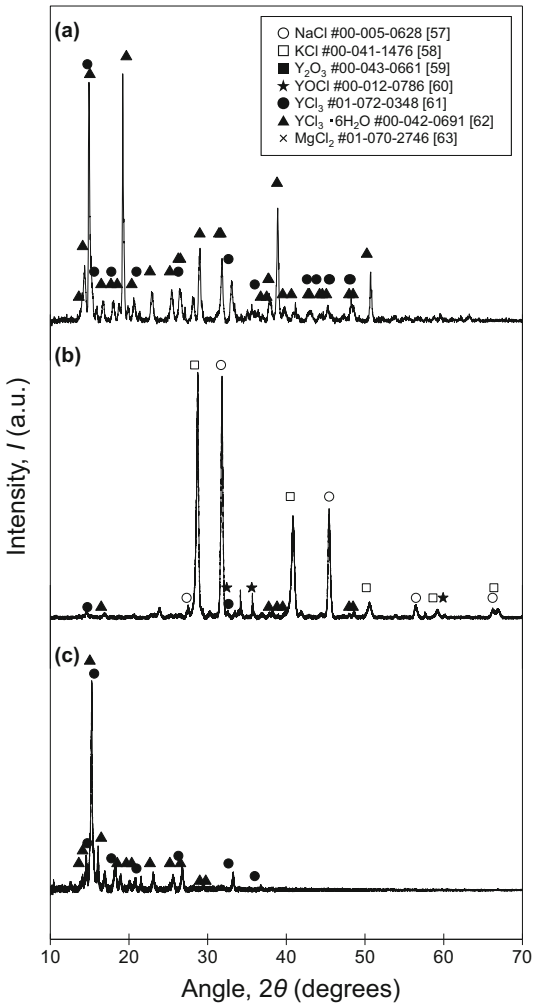


Fig. 6—XRD spectrum of the salts in the Ti crucible obtained after the experiments (1300 K (1027 °C), 173 ks). (*a*) After Exp. #181001-lot.3-6, (*b*) After Exp. #181001-lot.3-7, and (*c*) After Exp. #181001-lot.6-1[57–63]

contamination from Y metal can be prevented by the cyclic use of high-purity Y metal, which can be produced from YOCl during the N-free regeneration process (see the process flow shown in Figure 7).

The novel method developed in this study can deoxidize Ti scrap and Ti powder to an extremely low level using Y metal, which is oversupplied with the increasing demand of RE magnets. In addition, this process ensures economical rationality from the view point of the processing cost. For instance, 1-kg high-purity Ti material (*e.g.*, USD 20 to 50[64]) can be produced from Ti scrap with 10,000 ppm O by using approximately 40-g Y metal (*e.g.*, USD 1 to 5[65]). Furthermore, Figure 7 shows the flow chart of the proposed process combined with the regeneration of Y and YCl_3 from YOCl *via* electrolysis. When Y metal is recycled, this circular process does not consume the Y deoxidizing agent in principle. This study can be expected to establish a new industrial process that can reduce the O concentration in Ti more economically and reliably on a commercial scale, rather than the modification of existing industrial processes.

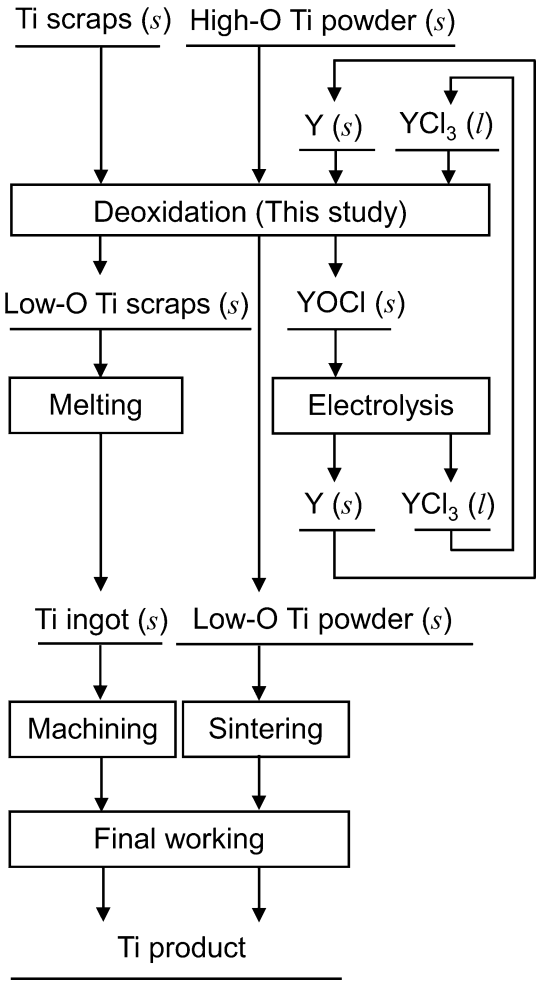


Fig. 7—A flowchart of applying the new deoxidation process using yttrium to the titanium industry.

V. CONCLUSION

In this study, we demonstrated a new deoxidation process of Ti using the Y/YOCl/YCl_3 equilibrium in molten chloride salt. The Ti samples were deoxidized, resulting in an O concentration of 30 to 60 ppm O in YCl_3 (*l*) at 1300 K (1027 °C). To the best of our knowledge, this is the first report of deoxidation technology, which produces Ti with extremely low O concentration using the RE/REOCl/$RECl_3$ equilibrium. This process will contribute to efficient recycling of Ti scrap and production of low O concentration Ti powder, thereby facilitating the application of Ti to the fields of aeronautics, robotics, and marine engineering.

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